

Lecture 37: Further variants of the loss function

Result: Unbiasedness & Bayes Estimator
No unbiased estimator $\delta(x)$ of $g(\theta)$ can be a Bayes estimator of $g(\theta)$ wrt the squared error loss except in a trivial situation.

Let $\delta(x)$ be an unbiased Bayes estimator of $g(\theta)$, under sq-error loss.

$\therefore \delta(x)$ is Bayes wrt squared error loss, $\therefore \delta(x) = E_{\theta|x}(g(\theta)|x=x)$

$\therefore \delta(x)$ is unbiased for $g(\theta) \therefore E(\delta(x)) = E_x E_{\theta|x}(g(\theta)|x=x) = g(\theta) \forall \theta$

$$\begin{aligned} \text{Consider, the jt. exp. } E_{x,\theta}[g(\theta)\delta(x)] &= E_{\theta} E_{x|\theta}[g(\theta)\delta(x)|\theta] = E_{\theta} g(\theta) E_{x|\theta}[\delta(x)|\theta] \\ &= E_{\theta} (g(\theta))^2 \quad (\because \delta(x) \text{ u.e. of } g(\theta)) \\ &= E_x E_{\theta|x}(g(\theta)\delta(x)|x) = E_x \delta(x) E_{\theta|x}(g(\theta)|x=x) \\ &= E_x (\delta(x))^2 \quad (\because \delta(x) \text{ Bayes est. of } g(\theta)) \end{aligned}$$

Hence, the risk in estimating $g(\theta)$ by $\delta(x)$

$$E_{x,\theta}(g(\theta) - \delta(x))^2 = E(g(\theta))^2 + E(\delta(x))^2 - 2E(g(\theta)\delta(x)) = 0$$

irrespective of the prior used here
only condition was squared error loss fn.
It is a trivial situation when $\delta(x) = g(\theta)$ w.p. 1.

Example

$X_1, \dots, X_n$ iid $N(\theta, \sigma^2)$, σ^2 known

$g(\theta) = \theta$ ($\mathbb{R}$), $L(\theta, d) = (\theta - d)^2$

Now $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ is an unbiased estimator of θ .

If $\bar{X}_n$ is the Bayes estimator, then the risk is

$$\begin{aligned} R(\theta, \bar{X}_n) &= E_{\theta} [E_{x|\theta}(\bar{X}_n - \theta)^2] = E_{\theta} \left(\frac{\sigma^2}{n} \right) = 0 \quad (\text{by the above result}) \\ &\Rightarrow \sigma^2 = 0 \Rightarrow \bar{X}_n \text{ is degenerate at } \theta. \end{aligned}$$

Some implications from the Normal-Normal set-up.

$X|\theta \sim N(\theta, \sigma^2)$, σ^2 known, $\theta \sim N(\mu_0, \sigma_0^2)$, so that $\theta|x \sim N(\eta, \tau^2)$,
where $\eta = \frac{n\bar{x}/\sigma^2 + \mu_0/\sigma_0^2}{n/\sigma^2 + 1/\sigma_0^2}$, $\tau^2 = \frac{1}{n/\sigma^2 + 1/\sigma_0^2}$ and wrt sq-error loss
 $\delta_{\text{Bayes}} = \eta(x)$

(i) No data case

$\delta_{\text{Bayes}} = \mu_0$ (the prior mean)

(ii) $\sigma_0^2 \rightarrow \infty$, any μ_0 (prior info' useless)

$\delta_{\text{Bayes}} \rightarrow \bar{x}$, the UMVUE (no value of the prior)

(iii) $\sigma_0^2 \rightarrow 0$, any μ_0 (prior says $\theta = \mu_0$ a.e. w.p. 1)

$\delta_{\text{Bayes}} \rightarrow \mu_0$ (full dependence on prior)

(iv) $\sigma^2 \rightarrow \infty$ (the conditional distn. useless)

$\delta_{\text{Bayes}} \rightarrow \mu_0$ (again, full faith on prior)

The Absolute Error Loss function

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$$L(\theta, d) = |\theta - d|$$

Bayes estimator minimizes the posterior risk, i.e.,

$$E[|\theta - \delta(x)| | X=x] \text{ which happens when } \delta(x) = \text{median of } \pi(\theta|x)$$

ex $x_1, \dots, x_n$ iid $\text{Ber}(\theta)$, $\theta \in [0, 1]$, $x = \{0, 1\}$.

$\theta \sim \text{Beta}(\alpha, \beta)$.

(Recall the $X \sim \text{Bin}(n, \theta)$, $\theta \sim \text{Beta}(\alpha, \beta)$ case)

here, δ_{Bayes} wrt sq. error loss is $\frac{\alpha + \sum x_i}{n + \alpha + \beta}$. came from

the posterior distn $\pi(\theta|x) \equiv \text{Gamma}(\alpha + \sum x_i, n + \beta - \sum x_i)$.

Now, if loss is absolute error, then δ_{Bayes} is the median of $\pi(\theta|x)$ i.e., we need to solve for δ (numerically) from

$$\frac{1}{B(\alpha + \sum x_i, n + \beta - \sum x_i)} \int_0^\delta \theta^{\alpha + \sum x_i - 1} (1 - \theta)^{n + \beta - \sum x_i - 1} d\theta = 0.5$$

(use tables of incomplete Beta fn $I_p(a, b) = \frac{1}{B(a, b)} \int_0^p u^{a-1} (1-u)^{b-1} du$)

The 0-1 loss function

$$L(\theta, d) = \begin{cases} 1 & \text{if } |\theta - d| > c \\ 0 & \text{if } |\theta - d| \leq c \end{cases}$$

To obtain Bayes estimator, we need to minimize posterior risk, i.e.,

$$\text{minimize } E[L(\theta, \delta(x)) | X=x] = P[|\theta - \delta(x)| > c | X=x] = 1 - P[|\theta - \delta(x)| \leq c | X=x],$$

i.e., maximize $P[\delta(x) - c \leq \theta \leq \delta(x) + c | X=x]$ (*)

i.e., choose an interval of length $2c$ which gives the maximum prob (*) or minimum $(1 - \text{prob}(*))$.

- Note
1. Such an interval is called modal interval of length $2c$.
 2. The Bayes estimate is the mid point of the modal interval of length $2c$.
 3. If $c \rightarrow 0$, the Bayes estimate is the posterior mode, i.e.,

$$\lim_{c \rightarrow 0} \delta_{\text{Bayes}}(c) = \delta^*, \text{ post posterior mode}$$